Track 1 report: *BUB1B* compound-heterozygous candidate in the MVA Hackathon 2026 proband

Participant: Binukumar (Hugging Face username) | **Repository:** https://github.com/drbinukumar/mva-hackathon-2026 (private until the hackathon ends; code is released under CC BY 4.0 as required) **Proband ID (dataset):** WGS_EX2312012 | **Genome build:** GRCh38

1. Summary

I prioritised variants in the organiser-provided whole-genome data using phenotype-driven ranking and a curated list of chromosomal-instability genes. The top candidate is a prioritised pair of variants in *BUB1B*, a mitotic-checkpoint gene linked to mosaic variegated aneuploidy(Hanks et al., 2004; Rio Frio et al., 2010). They form a potential compound heterozygote, but this is a candidate, not a confirmed genotype or diagnosis:

Allele	**Position (GRCh38)**	**Change**	**Consequence**	**Evidence summary**
1	chr15:40209701 T>G	NM_001211:c.2210T>G, p.Leu737Ter	stop-gained	Likely Pathogenic (ACMG/AMP 2015: PVS1, PM2; PP5 noted); predicted to trigger nonsense-mediated decay

Allele	Position (GRCh38)	Change	Consequence	Evidence summary
2	chr15:40220612 T>G	NM_001211:c.3006T>G, p.Asn1002Lys	missense (last exon)	VUS (ACMG/AMP 2015: PM2, PP3)

Important limitation: the two variants lie 10.9 kb apart and could not be phased with the data available (WhatsHap could not link them; no parental data). The trans (compound-heterozygous) configuration is therefore a hypothesis, not a confirmed finding. One incidental pharmacogenomic variant (*DPYD* rs1801160) is reported as a secondary finding with very low EPCR.

2. Predictions submitted

Rank	Variant(s)	finding_type	EPCR
1	chr15:40209701 T>G + chr15:40220612 T>G (*BUB1B*)	primary	0.4832
2	chr1:97305364 C>T (*DPYD*)	secondary	0.001

Only the ranking matters for scoring. EPCR values are computed by scripts/epcr/compute_epcr.py as a transparent heuristic (see section 4), not as calibrated probabilities.

3. Methods

Input. The organiser-provided WGS variant calls (GRCh38) and the proband's HPO terms from the phenotype document.

Annotation and ranking.

1. ANNOVAR(Wang et al., 2010) annotation using refGene, avsnp151, dbNSFP 4.7a(Liu et al., 2020; Wang et al., 2010), ClinVar (2025-04-09) (*Website*, n.d.), gnomAD 4.1(S. Chen et al., 2024), 1000 Genomes (Aug 2015), ExAC and ESP6500.
2. Phenotype-driven ranking with Exomiser 14.0.0(Smedley et al., 2015) (data release 2406) using the proband's HPO terms. *BUB1B* ranked first under the recessive model in the HPO-based run (gene combined score 0.967; phenotype score 0.784, variant score 0.926). The dominant-model entry for the same gene ranked third (combined score 0.867). Among the 27 panel genes, only *BUB1B* and *FANCD2* (recessive model, rank 4, combined score 0.861) had qualifying variants in this run.
3. First-pass filtering for rarity (allele frequency below 5% in 1000 Genomes and gnomAD 4.1 exomes and genomes) and for ClinVar pathogenic assertions, predicted loss of function or computational evidence of a damaging missense effect (criteria below), and a targeted check of a curated list of 27 chromosomal-instability and cancer-predisposition genes (Appendix A), followed by a genome-wide review for other genes carrying two or more qualifying variants.
4. Variants were classified manually against ACMG/AMP criteria.

Prioritisation criteria (scripts/variant_filtering/run_filtering.sh). (i) Retain exonic, UTR and splicing variants, excluding synonymous and unknown exonic changes, with allele frequency below 5% in 1000 Genomes, gnomAD 4.1 exomes and gnomAD 4.1 genomes (a first-pass screening threshold, not a disease-specific cut-off; missing values pass). (ii) Retain variants meeting at least one of: a ClinVar pathogenic or likely pathogenic assertion; a predicted

loss-of-function consequence (frameshift insertion or deletion, stop-gain, start-loss, splicing); or computational evidence of a damaging missense effect, defined as at least two of SIFT score <= 0.05, PolyPhen-2 HVAR score >= 0.447 and CADD PHRED >= 20 (used as PP3). (iii) Remove variants with benign or conflicting ClinVar classifications, flag variants absent from all population databases and ClinVar as novel, and list homozygous variants for recessive screening. (iv) Extract candidates in the 27-gene panel. (v) Separately, screen all intronic variants in the annotation file, without an allele-frequency filter, for a regSNPs-intron "disease-causing" prediction(Lin et al., 2019) , then restrict them to the panel genes. Candidates were then reviewed against the Exomiser phenotype ranking and classified manually under ACMG/AMP 2015 (section 4).

Intronic screen result. No intronic variant in the 27 panel genes, including *BUB1B*, was predicted disease-causing by regSNPs-intron. The screen covers single-nucleotide intronic variants that the annotation database scores; the absence of a prediction does not exclude deep-intronic, regulatory or splicing effects.

Supporting analyses.

- *Phasing:* WhatsHap(Martin et al., 2023) read-backed phasing; the two *BUB1B* variants could not be linked.
- *Structural variants:* Manta 1.6.0(X. Chen et al., 2016; Martin et al., 2023) calls were generated and calls overlapping the gene list were inspected; no structural variant was prioritised.
- *Mosaic aneuploidy screen:* a per-chromosome read-depth screen with mosdepth (Pedersen & Quinlan, 2018)(section 5).
- *Read-level inspection:* visual inspection of the proband's BWA (Li & Durbin, 2009)-aligned BAM in IGV 2.18.2(Robinson et al., 2011) at both candidate positions (section 5).

- *Incidental screen:* a screen of ACMG secondary-findings genes and pharmacogenomic variants, kept separate from the primary analysis.
- *Regulatory prediction:* AlphaGenome(Cheng et al., 2026) for the missense variant (section 5).

Software and databases. ANNOVAR, Exomiser 14.0.0, Manta 1.6.0, GATK 4.3.0.0, WhatsHap, mosdepth, AlphaGenome API 0.10.0, Snakemake 9.26.1, Python 3.11. Exact environment details are in environment.yml; entries that are unpinned there were not recorded at run time.

Tools and resources used: ANNOVAR(Wang et al., 2010), Exomiser(Smedley et al., 2015), Manta (X. Chen et al., 2016), mosdepth(X. Chen et al., 2016; Pedersen & Quinlan, 2018), WhatsHap(Martin et al., 2023), ClinVar (*Website*, n.d.), gnomAD 4.1(S. Chen et al., 2024), dbNSFP(Liu et al., 2020), AlphaMissense(Cheng et al., 2023; Liu et al., 2020), regSNPs-intron annotations in ANNOVAR, AlphaGenome, IGV and BWA.

4. Evidence for the two *BUB1B* alleles

Variants were classified manually under the original ACMG/AMP 2015 guidelines, using the published criteria strengths and combining rules (Richards et al., 2015). Population frequencies are from gnomAD 4.1 as annotated by ANNOVAR, ClinVar assertions are from the 2025-04-09 release, and in-silico scores are from dbNSFP 4.7a. The ACMG/AMP criteria assigned to each allele are listed below.

Allele 1 (stop-gained, c.2210T>G, p.Leu737Ter): criteria PVS1, PM2 and PP5; classification Likely pathogenic. *PVS1 (very strong):* nonsense variant in exon 17 of 23 of a gene in which loss of function is the established disease mechanism (biallelic *BUB1B* loss of function causes mosaic variegated aneuploidy syndrome 1). The stop lies 747 nt upstream of the last exon-exon junction, beyond the roughly 50-55 nt limit, so nonsense-mediated decay (NMD) is predicted. *PM2 (moderate in the 2015 guidelines):* very rare in population data (gnomAD 4.1 exome allele frequency 7.9 x 10^-5; genome 3.3 x 10^-5) . *PP5 (supporting, as defined in 2015):*

ClinVar lists pathogenic and likely pathogenic assertions for this variant from several submitters, and the variant is discussed in the *BUB1B* literature. Under the 2015 combining rules, PVS1 plus one moderate criterion (PM2) gives Likely pathogenic. PP5 is supporting context from ClinVar assertions; it is not counted independently of the evidence behind those assertions, and if it were counted the 2015 rules would reach Pathogenic, so the more conservative Likely pathogenic classification is reported. Later ClinGen recommendations downgrade PM2 to supporting strength and retire PP5; under them PVS1 plus PM2 (supporting) is 9 points on the Tavtigian scale(Tavtigian et al., 2018), still likely pathogenic, so the conclusion does not depend on the framework version.

Allele 2 (missense, c.3006T>G, p.Asn1002Lys): criteria PM2 and PP3; classification variant of uncertain significance (VUS). *PM2 (moderate in the 2015 guidelines):* extremely rare (gnomAD 4.1 exome allele frequency 6.8 x 10^-7; not observed in the genome data). *PP3 (supporting):* several computational predictors indicate a damaging effect (SIFT deleterious, PolyPhen-2 probably damaging, CADD 24.5, AlphaMissense 0.923, MVP 0.852). These predictors are correlated and were counted once. REVEL (0.472) is indeterminate and was not counted as supporting evidence. The variant lies in the last exon (exon 23), where NMD does not apply, so any effect would be at the protein level. A different nucleotide change causing the same amino-acid change (c.3006T>A) is listed in ClinVar as a VUS; this is context only and is not a classification of c.3006T>G. PM2 plus PP3 does not satisfy any 2015 combining rule for Likely pathogenic or Pathogenic, so the variant remains a VUS.

EPCR calculation. ACMG/AMP evidence is converted to points on the Tavtigian et al. scale (very strong 8, strong 4, moderate 2, supporting 1) and then to a posterior probability (prior 0.1, odds of pathogenicity 350^(points/8))(Tavtigian et al., 2018). Allele 1 scores 11 points (PVS1 8, PM2 2, PP5 1) and allele 2 scores 3 points (PM2 2, PP3 1). For a compound-heterozygous pair the weaker allele is used (both alleles are required), and the result is scaled by the Exomiser gene-level combined score (0.967 for *BUB1B*). Secondary findings are not scaled by the gene score. The ACMG criteria were assigned by hand; only the conversion is scripted. The pair

therefore scores 0.4832 (weaker-allele posterior about 0.50 x 0.967), and the incidental *DPYD* variant is clamped to 0.001. Under the later ClinGen strengths (PM2 supporting, PP5 not used) allele 2 would have 2 points and the pair would score about 0.31; the ranking would not change. EPCR is a heuristic prioritisation score, not an ACMG classification or a calibrated clinical probability.

Secondary finding. *DPYD* c.2194G>A (p.Val732Ile, rs1801160, DPYD*6) has ClinVar annotation "drug response" and a population allele frequency of about 4-5%. It is an incidental pharmacogenomic observation unrelated to the proband's phenotype and is not claimed as causal.

5. Exploratory analyses (descriptive, not validated calls)

IGV read-level inspection. The proband's sorted BAM was inspected in IGV 2.18.2 at both candidate positions (counts read from the IGV coverage pop-up). At chr15:40209701 T>G, 27 of 52 reads carried G (52%; 15 forward, 12 reverse) and 25 carried the reference T (14 forward, 11 reverse). At chr15:40220612 T>G, 16 of 32 reads carried G (50%; 6 forward, 10 reverse) and 16 carried T (5 forward, 11 reverse). Alternate-allele support was present on both strands at both sites, consistent with heterozygous variants. This is a visual check of read support only; mapping quality, base quality and local alignment artefacts were not quantified, and the observation does not show that the two variants are on different haplotypes.

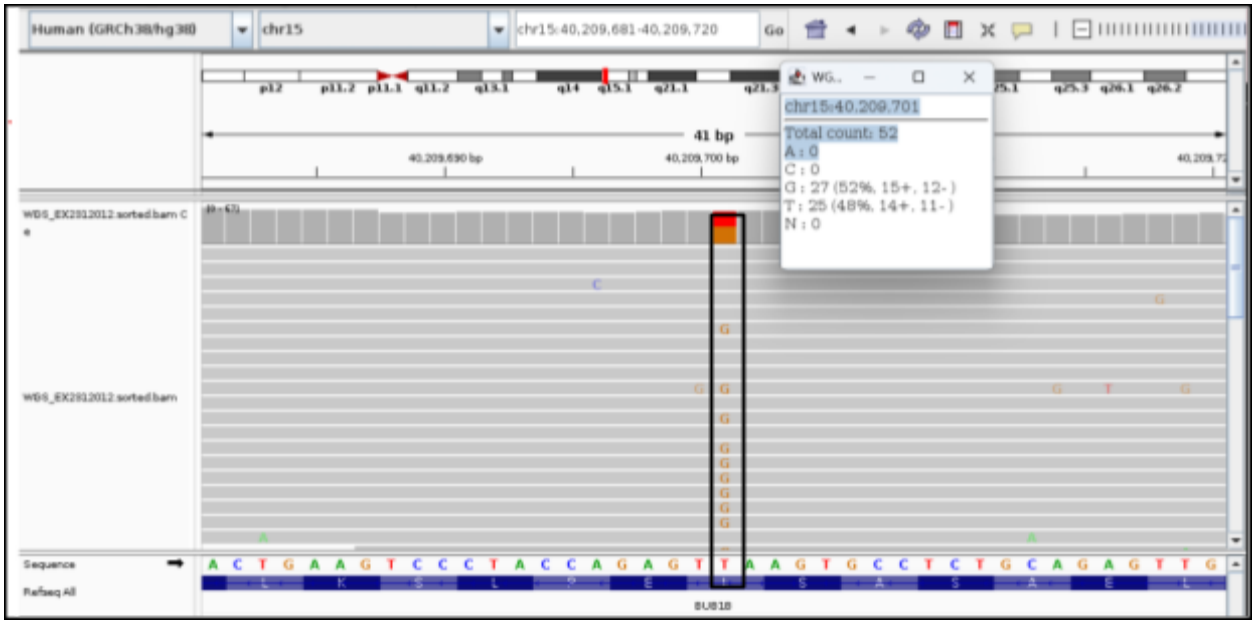


Figure 1. *IGV view of the proband's alignments at chr15:40209701 (BUB1B c.2210T>G, p.Leu737Ter): 27 of 52 reads carry G (15 forward, 12 reverse).*

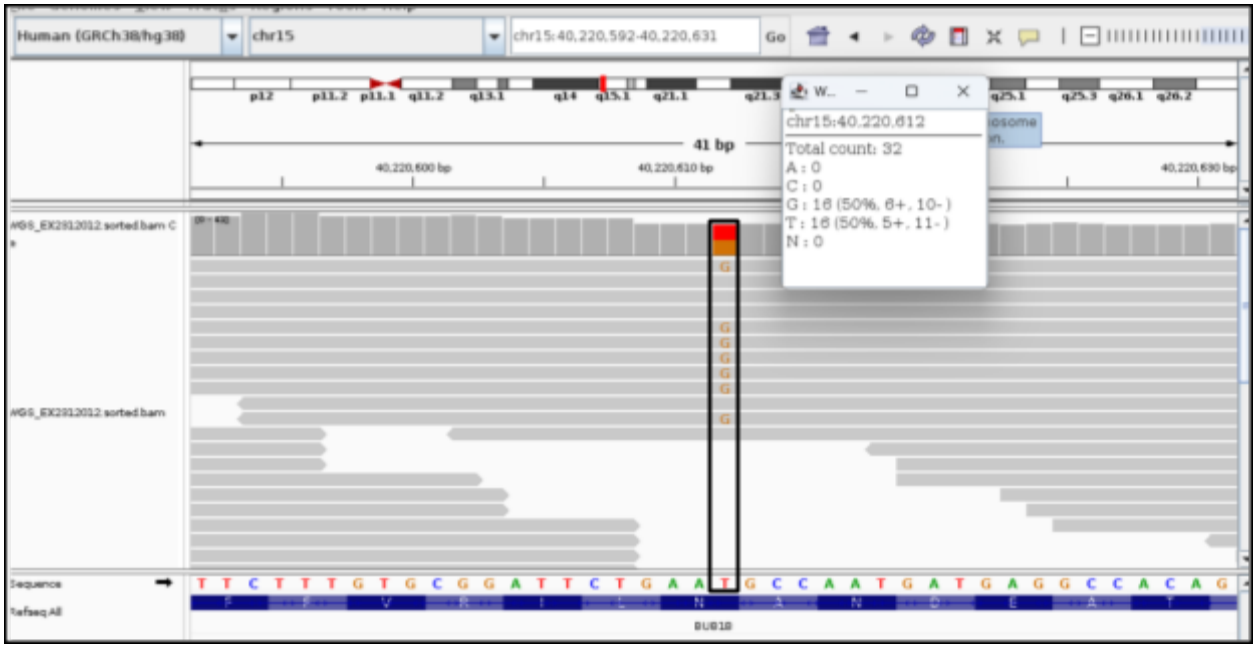


Figure 2. *IGV view of the proband's alignments at chr15:40220612 (BUB1B c.3006T>G, p.Asn1002Lys): 16 of 32 reads carry G (6 forward, 10 reverse).*

Per-chromosome read depth (mosdepth, 1 Mb windows). Autosomal depth ratios (median window depth relative to the autosomal median) were mostly between 0.986 and 1.010. Small elevations on chromosomes 16, 17, 19 and 22 (ratios 1.030 to 1.051) are in the most GC-rich chromosomes and follow the typical GC-bias pattern; a mosaic trisomy would be unlikely to affect four GC-rich chromosomes while leaving others flat. Chromosomes X and Y had about half of the autosomal depth (0.49 and 0.51), consistent with one copy of each. Bulk depth in a single sample has limited sensitivity to low-level or cell-type-restricted mosaicism, so this result does not exclude mosaic aneuploidy.

AlphaGenome regulatory prediction (missense variant). AlphaGenome API v0.10.0 (non-commercial use) was run over a 1 Mb window around chr15:40220612 T>G. Predicted RNA-seq signal across the *BUB1B* locus was essentially unchanged between reference and alternate alleles (ALT/REF = 0.9994). The tissue term used (UBERON:0001157) was taken from the tool's quickstart and is not matched to the proband's cell type, and only RNA-seq output was requested. This result says nothing about the pathogenicity of the amino-acid change and did not alter classification or ranking. AlphaGenome output is subject to its non-commercial terms of use.

6. Limitations

- Single case; no parental data, so the trans configuration of the two *BUB1B* variants is unconfirmed.
- Allele 2 is a VUS; the assignment of the pair as causal rests on allele 1, the phenotype and the gene's disease association.
- Read-level support for the two variants was checked visually in IGV only; mapping quality, base quality and alignment artefacts were not quantified.
- The mosaic-aneuploidy screen is exploratory and has limited sensitivity.
- Repeat expansions were not performed.

- ACMG/AMP evidence assignments were performed manually. Exomiser was used for variant prioritisation; its prioritisation scores were treated as supporting evidence rather than as a substitute for ACMG/AMP classification
- PM2 relies on gnomAD frequencies; site-level data quality at the two positions was not independently audited. The original ACMG/AMP 2015 strengths were used (PM2 moderate, PP5 applied); later ClinGen recommendations differ (see section 4).
- The intronic screen relies on regSNPs-intron predictions in ANNOVAR (single-nucleotide intronic variants only), so intronic indels and unscored or regulatory variants were not assessed.
- EPCR values are heuristic and uncalibrated.

7. Suggested follow-up

Targeted testing of both variants in the parents through a clinical laboratory would show whether each was inherited from a different parent. Long-read sequencing of the proband could phase the two variants directly. A plain-language summary is in docs/family_summary.md. This report is research analysis and is not medical advice.

8. Use of AI assistance

- **Provider and model:** Anthropic Claude (Sonnet 5.5)
- **Access and plan:** claude.ai consumer chat, Free plan
- **Data-handling setting:** "Help improve Claude" (use of chats for model training) was **off**
- **Use:** help with command-line usage and script drafting, checking interpretation, and drafting this report. Helper scripts in the repository were drafted with Claude; I ran them and checked their outputs. All variant calls, annotations and depth results come from the pipeline I ran on institutional HPC, and I reviewed and decided the final candidates and EPCR assignments.
- **Data entered into the assistants:** excerpts of annotation output, filtered candidate-variant tables (Exomiser and ANNOVAR rows for the prioritised genes and variants) and the repository's scripts and summary files were shared with the assistants

during analysis and review. No raw reads, alignments or complete unfiltered variant files were uploaded.

- AlphaGenome (section 5) is a prediction tool, not an assistant, and is listed under methods.

9. Privacy

This report and the repository contain no raw reads, alignment files, complete variant files, or information that identifies the patient or family beyond what the organisers have already published. Analyses used the organiser-provided data under the hackathon data-use terms.

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Appendix A. Gene list used for panel filtering (27 genes)

The list was curated by the participant using OMIM, GeneReviews and the published literature shown below. Tier 1: mosaic variegated aneuploidy (MVA) genes and spindle-assembly checkpoint genes; Tier 2: DNA-repair and chromosome-breakage syndrome genes; Tier 3: cancer-predisposition genes relevant to the proband's phenotype (including rhabdomyosarcoma). It was used to restrict the first-pass filter; a genome-wide review for other genes with qualifying variants was also performed. Entries are as recorded by the participant.

Gene	**Tier**	**OMIM gene**	**OMIM disease**	**Locus**	**Key literature / note**
BUB1B	1	602860	257300 (MVA1)	15q15.1	Hanks 2004; Rio Frio 2010; Ochiai 2004; Bohers 2008; Schmid 2014; Ozono 2021

Gene	**Tier**	**OMIM gene**	**OMIM disease**	**Locus**	**Key literature / note**
CEP57	1	607951	614114 (MVA2)	11q21	Snape 2011; Pinson 2014
TRIP13	1	604507	617598 (MVA3)	5p15.33	Yost 2017
CENATAC	1	620142	620153 (MVA4)	11q23	OMIM
SLF2	1	610348	620184 (MVA5)	10q24.31	OMIM
SMC5	1	609386	620185 (MVA6)	9q21.12	PMC12691382, 2025
MAD1L1	1	602686	620189 (MVA7)	7p22.3	Villarroya-Beltri 2022
BUB1	1	602452	Associated	2q13	Matsuura 2006
BUB3	1	603719	Associated	10q26.13	Mechanistic
NBN	2	602667	251260 (NBS)	8q21	Wu 2016
FANCA	2	607139	227650	16q24	GeneReviews
FANCC	2	613899	227645	9q22.32	OMIM

Gene	**Tier**	**OMIM gene**	**OMIM disease**	**Locus**	**Key literature / note**
FANCG	2	602956	614082	9p13.3	de Winter 1998
FANCD2	2	613984	227646	3p25	Exomiser rank #4
BLM	2	604610	210900 (Bloom)	15q26	German 1994
ATM	2	607585	208900 (A-T)	11q22.3	Savitsky 1995
ESCO2	2	609353	268300 (Roberts)	8p21.1	Vega 2005/2010
TP53	3	191170	151623 (Li-Fraumeni)	17p13	Perrino 2021 (n=11/45)
NF1	3	613113	162200	17q11.2	Perrino 2021 (n=9/45)
HRAS	3	190020	218040 (Costello)	11p15.5	Perrino 2021
DICER1	3	606241	601200 (PPB/DICER 1 syndrome)	14q32.13	Perrino 2021

Gene	**Tier**	**OMIM gene**	**OMIM disease**	**Locus**	**Key literature / note**
PTCH1	3	601309	109400 (Gorlin/BCNS1)	9q22.32	Perrino 2021
SUFU	3	607035	155255 (Medulloblastoma); 109400 (Gorlin)	10q24.32	Perrino 2021; Taylor 2002
BRCA2	3	600185	605724 (FANCD1)	13q13.1	Perrino 2021; Reid 2005 (medulloblastoma)
CHEK2	3	604373	609265 (Tumor Predisposition Syndrome 4, not classic Li-Fraumeni)	22q12.1	Bell 1999
LIG4	3	601837	606593 (LIG4 syndrome)	13q33	O'Driscoll 2001; phenotype resembles NBS
MEN1	3	613733	131100	11q13	Ozono 2021

Appendix B. Methods description (answers to the organisers' template)

Team name. Binukumar

Model number. 1

Model/approach in detail. Phenotype-driven prioritisation of the organiser-provided whole-genome (WGS) data for the proband (GRCh38). The organiser VCF was annotated with ANNOVAR (refGene, avsnp151, dbNSFP 4.7a, ClinVar 2025-04-09, gnomAD 4.1, 1000 Genomes Aug 2015, ExAC, ESP6500) and independently ranked with Exomiser 14.0.0 (data release 2406) using the proband's HPO terms. Variants were filtered for rarity (AF < 5% in 1000 Genomes and gnomAD) and predicted impact, checked against a curated list of 27 chromosomal-instability and cancer-predisposition genes, and followed by a genome-wide review for genes with two or more qualifying variants. BUB1B ranked first in the HPO-only Exomiser run (gene combined score 0.967). Supporting analyses: WhatsHap read-backed phasing, Manta 1.6.0 structural variants, and a per-chromosome read-depth screen for mosaic aneuploidy (mosdepth, 1 Mb windows).

Generative AI use (provider, plan, setting). Anthropic Claude (Sonnet 5.5), claude.ai consumer chat, Free plan. Setting 'Help improve Claude' (training on chats) was OFF. Used for command and script help and report drafting. All variant calls and annotations were produced by the pipeline run on institutional HPC; the participant reviewed and decided the final candidates.

Automated output or downstream manual review?. Partly automated: ACMG/AMP evidence criteria were assigned manually; EPCR values were computed by script (scripts/epcr/compute_epcr.py) from ACMG points and the Exomiser gene score.

Downstream manual review. Candidates were selected by manual review of the Exomiser and ANNOVAR outputs and classified by hand against ACMG/AMP criteria. EPCR was then

computed by script: ACMG points converted with the Tavtigian Bayesian point framework, weaker allele used for a compound-het pair, scaled by the Exomiser gene combined score (0.4832 for the BUB1B pair; 0.001 for the incidental finding).

Public or proprietary data. Organiser-provided challenge WGS data (gated hackathon dataset) together with publicly available reference and annotation resources. No additional proprietary external datasets were used.

Public data used. Reference genome GRCh38. Annotation: ANNOVAR refGene, avsnp151, dbNSFP 4.7a (SIFT, PolyPhen2, MVP, CADD, REVEL, AlphaMissense), ClinVar 2025-04-09, gnomAD 4.1 exomes/genomes, 1000 Genomes Aug 2015, ExAC, ESP6500. Phenotype ranking: Exomiser 14.0.0 with data release 2406 (HPO, gene-phenotype resources). Literature and gene-disease knowledge on mosaic variegated aneuploidy genes for the gene list. AlphaGenome API 0.10.0 (non-commercial terms) was used for an exploratory regulatory prediction on the missense variant. Gene list of 27 chromosomal-instability and cancer-predisposition genes curated by the participant (see report, Appendix A).

Proprietary data used. No additional proprietary external datasets were used. The primary WGS data were provided by the hackathon organisers under the data-use terms.

Compound-heterozygous pairs or single variants?. Both. The submission contains one compound-heterozygous pair (two variants in one row, using the _2 columns). Phase could not be confirmed (WhatsHap could not link the two variants, 10.9 kb apart), so the pair is reported as a candidate rather than a confirmed trans configuration.

Secondary / incidental findings. Common incidental and pharmacogenomic variants were screened separately from the main analysis. Only one incidental observation, *DPYD* rs1801160 (p.Val732Ile), was included; it is flagged finding_type=secondary with a very low EPCR (0.001) and a note that it is not a causal claim.

Run time and cost. Run on institutional HPC (IGIB Tejas, Slurm); no cloud cost. Analysis ran over several weeks; exact total compute time was not tallied.

Method abstract (strengths and limitations). We prioritised variants in the organiser WGS data using phenotype-driven ranking. The variant file was annotated with ANNOVAR using public databases and ranked with Exomiser 14.0.0 from the proband's HPO terms. Candidates were restricted to rare variants (AF < 5% in the first-pass screen) with predicted functional impact and compared against a curated list of 27 chromosomal-instability and cancer-predisposition genes, followed by a genome-wide check for other genes with two or more qualifying variants. *BUB1B* ranked first in the HPO-only run (gene combined score 0.967). Two BUB1B variants were prioritised as a candidate compound-heterozygous pair: a stop-gained variant (c.2210T>G, p.Leu737Ter; ClinVar pathogenic/likely pathogenic; predicted to trigger nonsense-mediated decay) and a missense variant (c.3006T>G, p.Asn1002Lys; VUS under ACMG/AMP). EPCR was computed by script from hand-assigned ACMG points (Bayesian point framework), using the weaker allele of the pair and scaling by the Exomiser gene score. Supporting analyses included WhatsHap phasing, Manta structural-variant calling, a per-chromosome read-depth screen for mosaic aneuploidy, and an exploratory AlphaGenome regulatory prediction. Strengths: public annotation resources and organiser-provided challenge data, phenotype-aware ranking, a genome-wide review beyond the gene list, reproducible EPCR calculation, and transparent reporting of uncertainty. Limitations: the two BUB1B variants could not be phased (no parental data), so the trans configuration is unconfirmed; the missense allele is a VUS; the

mosaicism screen is exploratory with limited sensitivity; Read-level support was assessed visually in IGV; mapping quality, base quality and local alignment artefacts were not quantified.; repeat expansions were not assessed; ACMG assignments were manual. Single-case analysis.